Fiscal Year:	FY 2009	**Task Last Updated:**	FY 02/17/2009
PI Name:	Nalette, Timothy Ph.D.		
Project Title:	Advanced CO2, Humidity, and Trace Contaminant Control System for Closed Loop Life Support Systems		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	INDUSTRY	**Phone:**	860-654-2403
Organization Name:	Hamilton Sundstrand Space Systems International, Inc.		
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PI Address 2:			
PI Web Page:			
City:	Windsor Locks	**State:**	CT
Zip Code:	06096-1010	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	03-OBPR-01
Start Date:	08/11/2004	**End Date:**	10/31/2008
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 10/31/2008 per PI (10/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Papale, William (Hamilton Sundstrand)		
Grant/Contract No.:	NNJ04HF73A		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Conduct basic research associated with amine based sorbents for combined humidity and carbon dioxide control for closed loop life support. The basic research will investigate alternate sorbent formulations and configurations that will address enhanced kinetics and loadings to enhance overall system performance. These enhancements will result in reduced system mass, volume and power, while also enhancing overall vehicle integration issues associated with the addition of the humidity control function. Based on the results of the basic sorbent research, we will design, fabricate, assemble and test a full scale, deliverable prototype system.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	May have applicability to CO2 capture from stationary power plants to address greenhouse gas abatement.
Task Progress:	On August 11, 2004, Hamilton Sundstrand Space Systems International (HSSSI) was awarded a cooperative agreement (NNJ04HF73A) to pursue the development of an Advanced CO2, Humidity, and Trace Contaminant Control System for Closed Loop Life Support Systems Project. A Rapid Technology Development Team consisting of Hamilton Sundstrand, The University of Hartford, The University of Connecticut, and NASA implemented a four year effort to develop prototype hardware capable of removing CO2, H2O, and trace contaminants from a cabin atmosphere. The team's original focus was the development of a regenerable, amine-based system, which combines CO2, humidity, and trace contaminant control (TCC) in a single system – however, at the conclusion of the first year of the contract the TCC component of the design was eliminated. At the conclusion of the program, testing of prototype systems will demonstrate a Technology Readiness Level (TRL) of TRL-6, evaluating full-scale system performance in a relevant environment. Hamilton Sundstrand (HS) developed an amine-based sorbent, designated SA9T, possessing an approximate two-fold increase in CO2 adsorption capacity relative to previous HS solid amine formulations. As such, SA9T was selected as the baseline sorbent for this program. In addition to CO2 capacity, SA9T also adsorbs water vapor and can easily be regenerated upon exposure to vacuum. This program investigates dual functioning sorbents as a combined humidity and CO2 control system, using SA9T as the baseline while concurrently investigating new sorbents being evaluated by University teammates. During the first year of the project, basic research was conducted on amine synthesis, and alternate amine support materials, performance and trace contaminant challenges were defined and preliminary testing of the material initiated. Early in the second year of the project, NASA redirected the program to focus on the design as it relates specifically to CEV. To this end, and to support the verification of the TRL of the amine based sorbent technology, the design phase of the program was accelerated, while the basic university research continued. The baseline sorbent, SA9T was selected, and due to the lower than desired trace contaminant removal efficiency, the trace contaminant removal component of the project was eliminated. Additionally the system design targeted a CEV application, and commonality with EVA use was not considered. Based on early CEV design input for planned crew sizes and system redundancy requirements, the sorbent canister was sized for a crew of six persons under emergency conditions in which only one system is operational. Under nominal conditions with two systems operating, the canister size will support a crew of three or more persons depending on metabolic load. The deliverable quantity of systems was increased to two, with the first delivery accelerated to June 2006. The second system delivery was scheduled for the last quarter of 2006, however, delivery was rescheduled to February 2007 to allow for design, fabrication and incorporation of sweep gas regeneration capabilities. Throughout 2006 and 2007, the third year of the effort, NASA-Johnson Space Center conducted numerous chamber tests to validate system performance in representative environments, as detailed in CO2 and Moisture Removal Amine Swing-bed (CAMRAS) test reports generated by Crew and Thermal Systems Division at NASA-JSC. Hamilton Sundstrand conducted isotherm testing on SA9T to evaluate CO2 and moisture capacity at different temperatures and CO2/H2O concentrations, as well as investigation of competitive adsorption between CO2 and water vapor. Following successful testing of each prototype system at NASA-Johnson Space Center and refinements in system operability requirements for CEV, design and construction of a third prototype system (CAMRAS Unit #3) was initiated in FY2008, the fourth year of the contract. The design of the third prototype system incorporated changes to the system flow diverter valve to improve system fluid interfaces and overall system packaging. A multi-ball valve assembly concept was selected over several alternative valve concepts. Furthermore, some changes to the canister design were necessary to accommodate the flow distribution to and from the new diverter valve. Subsequent to design efforts beginning in November 2007 and a Technical Interchange Meeting at NASA-JSC in February 2008, manufacture of the third system commenced in the third quarter of FY2008. The valve and canister assemblies were completed in October 2008 with delivery to NASA-JSC in November 2008 following post-assembly performance evaluations at Hamilton Sundstrand. In addition to the design and fabrication of the third prototype system during the fourth year of the contract effort, Hamilton Sundstrand conducted sub-scale evaluations on SA9T to investigate the sorbent capability in removing selected trace organic contaminants under similar operating conditions as the full-scale prototypes (similar residence time and half-cycle time). Results from this testing demonstrated that the solid amine sorbent SA9T has a cyclic capacity for some of the organic contaminants tested, namely methanol, ethanol and xylenes. The sorbent also demonstrated removal of aldehydes, such as formaldehyde and acetaldehyde, however, the reaction with aldehydes is not reversible.
Bibliography Type:	Description: (Last Updated: 02/19/2009)